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R18

Course Code: A30461



CMR COLLEGE OF ENGINEERING & TECHNOLOGY
(UGC AUTONOMOUS)

B.Tech III Semester Supplementary Examinations December-2024

Course Name: Analog & Digital Electronics

(Common for CSE & IT)

Date: 19.12.2024 AN

Time: 3 hours

Max.Marks: 70

(Note: Assume suitable data if necessary)

PART-A

Answer all TEN questions

Each question carries TWO marks.

10x2=20M

1. Draw the VI characteristics of ideal pn junction diode. 2 M
2. What is meant by avalanche breakdown? 2 M
3. Define early effect. 2 M
4. How transistor acts as an amplifier. 2 M
5. Differentiate between enhancement and depletion modes of MOSFET. 2 M
6. Define 'Minterm' and 'Maxterm'. 2 M
7. Convert $A3B_H$ and $2F3_H$ into Binary and Octal respectively. 2 M
8. How do you draw a full adder circuit? 2 M
9. Distinguish between combinational and sequential circuits. 2 M
10. List out the applications of shift registers. 2 M

PART-B

Answer the following. Each question carries TEN Marks.

5x10=50M

- 11.A). Give the diode current equation. Show that the Zener diode is used as voltage regulator. 10M

OR

11. B). Explain the operation of bridge rectifier using diode with neat waveforms. 10M

12. A). With a neat diagram, explain the input-output characteristics of common base configuration of NPN transistor. 10M

OR

12. B). What is called intrinsic stand-off ratio of UJT? Determine the frequency of oscillation of an UJT relaxation oscillator. Assuming $R_e = 10.7K\Omega$, $C_e = 0.22 \mu F$ and Intrinsic stand-off ratio = 0.56. Give the applications of UJT. 10M

13. A). Explain the construction and operation of JFET with neat sketches. 10M

OR

13. B). i) Find Excess-3 code for the following decimal numbers: $(18)_{10}$ and $(56)_{10}$. 4M
 ii) Identify the equivalent Gray code for $[10110]_2$; Add $(65)_{10} + (58)_{10}$. 3M
 iii) Perform 2's complement subtraction of $010110 - 100101$. 3M

14. A). Prepare Karnaugh map for $Y(A, B, C, D) = \prod M(0,3,4,9,10,12) + \prod d(2,7,8,13)$ and draw the logic diagram using basic gates. 10M

OR

14. B). Plot the following logical Expression on a 4-variable K - map 10M
 $F = ABCD + AB'C'D' + AB'C + AB$ & realize the SOP using only NAND gates and POS using only NOR gates.

(P.T.O..)

15. A). Illustrate the design procedure of a MOD-5 synchronous counter using JK flip-flops and implement it. 10M

OR

15. B). Design a synchronous sequential circuit using JK Flip-flop for the given state diagram. 10M


